Qi Wu

Optical Metrology · Surface Science · Instrumentation
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Profile. Researcher and hands-on engineer with four years at imec and KU Leuven working on the optical characterization of liquids and surfaces, and on building the instruments to measure them. Work spans surface modification and wetting, custom ATR-FTIR and contact-angle metrology, optical simulation, and data analysis in Python and MATLAB. Cross-disciplinary background across optics, electrical engineering, and MEMS.

EDUCATION

Ph.D. in Chemistry — *KU Leuven & imec, Belgium | 2022–2026 (expected)*

M.Sc. in Electrical Engineering — *University of Twente, the Netherlands | 2019–2021*

B.Eng. in Electrical Engineering — *Northwestern Polytechnical University, China | 2015–2019*

TECHNICAL SKILLS

Characterization & Metrology: FTIR spectroscopy; ellipsometry; wettability analysis; SEM; AFM.

Instrumentation & Prototyping: CAD design and build of custom metrology systems; re-engineering of commercial instruments; developing embedded microcontrollers; 3D-printing prototyping; measurement automation.

Optical Simulation: FDTD simulation of evanescent-field and near-field optics; spectral correction, processing, and analysis.

Programming & Data: Python and MATLAB for data processing, analysis, and hardware-software integration.

Experiment & Process: design of experiments (DOE); self-assembled monolayers (SAMs) LPCVD; wet-chemical surface processing; 2,000+ hrs cleanroom operation experience.

Languages: English (professional), Dutch (beginner), Mandarin Chinese (native).

RESEARCH EXPERIENCE

PhD Researcher — *imec, Leuven, Belgium | 2022–Present*

Superhydrophobic Surface Fabrication by Silane SAM Deposition

- **Surface engineering:** engineered superhydrophobic silicon surfaces via optimized silane self-assembled monolayer (SAM) deposition on patterned substrates, reaching intrinsic and apparent water contact angles above 110° and 140°.
- **Mechanistic analysis:** resolved the deposition mechanism by GATR-FTIR spectroscopy, profiling the kinetic balance between surface -OH loss during HF pre-cleaning and re-oxidation during CVD.
- **Process control:** established a robust pre-cleaning and deposition SOP with >90% run-to-run reproducibility; recognized with the Best Student Paper Award at UCPSS 2025.

Quantitative Optical Metrology of Wetting Defects in HAR Nanopatterns

- **Instrument development:** built a customized ATR module for a commercial FTIR platform with an adjustable optical path, tuning the number of IR reflections to raise signal-to-noise ratio while avoiding absorption saturation.
- **Structure-to-spectrum mapping:** established a quantitative relation between nanopattern geometry (height 50-800 nm, surface fraction 0.1-0.8, at fixed 90 nm pitch) and ATR-FTIR absorbance, yielding a forward model from which the air-trapped fraction is inverted from a measured spectrum.
- **Physical verification of the model:** the near-field distribution at the structured interface under total internal reflection was computed by FDTD, confirming that the measured absorbance change originates from the filled trench geometry.
- **Validity domain:** defined the geometric window where the proportionality between air-trapping rate and absorbance loss remains valid (height < 500 nm, surface fraction < 0.7); the two bounds were traced to the finite penetration depth and to breakdown of the total internal reflection condition.

Synchronized Cross-View Imaging Platform for Anisotropic Wetting Characterization

- **Platform design:** designed and built a cross-view optical platform for simultaneous, dual-directional imaging of sessile droplets on nano-trenches.
- **Measurement accuracy:** eliminated the timing mismatch of sequential imaging, enabling synchronized capture of anisotropic droplet elongation with improved morphological accuracy.

- **New defect indicator:** identified the structural-parameter regime where conventional wetting models break down, and established real-time anisotropic elongation as a new indicator of nanoscale wetting defects.

Graduate Intern *ARCNL, Amsterdam, the Netherlands | 2021*

- **Mechanical simulation:** ran stress-strain simulations (COMSOL) of elastomer microstructure arrays for low-wear wafer carriers.
- **Industrial collaboration:** worked with cleanroom specialists to improve micro-component durability in a research-to-manufacturing setting.

PUBLICATIONS & PRESENTATIONS

Publications

- Q. Wu et al., *"Modulating Silane Monolayer Grafting on Nanopatterned Silicon: Insights from Infrared-Spectroscopy-Based Air-Trapping Monitoring,"* **Solid State Phenomena** (UCPSS 2025, 17th International Symposium on Ultra Clean Processing of Semiconductor Surfaces), accepted & in press.

Presentations

- Q. Wu, *"Impact of HF Pretreatment on Silane Self-Assembled Monolayers on Silicon Surfaces,"* **oral presentation**, 18th Mini-Symposium on Liquids, Okayama, Japan, May 2025.
- Q. Wu, *"Optical Method to Characterize the Wetting States on Nano-patterned Surfaces,"* **poster presentation**, SPP 2171 Summer School, Munster, Germany, July 2024.

INTERESTS

Tennis, bouldering, hiking, DIY electronics.